

- Q.31 Write a comparison between LED and LCD.
 Q.32 How addition and subtraction can be performed with the help of operational amplifier.
 Q.33 Sketch the circuit diagram of Op-Amp as an integrator with explanation.
 Q.34 Explain the construction and working theory of seven segment display.
 Q.35 What are loading effects and how they can be minimized?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the working principle of X-Y recorder with its constructional details.
 Q.37 What are the factors to be considered while selecting an instrument?
 Q.38 Draw a pin diagram of an operational amplifier IC-741 with each pin explanation.

No. of Printed Pages : 4

181534/121534/031534

Roll No.

/073633

3rd Sem / IC, EI

Subject:- Basics of Instrumentation / Pr. Of Inst.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The process of measuring and control of desired quantity collectively is called as
 a) Measuring b) Sensing
 c) Instrumentation d) None
- Q.2 Galvanometer is an example of which type of instrument.
 a) Absolute b) Deflecting
 c) Reflecting d) None
- Q.3 The error in which the difference is taken between measured value and true value.
 a) Fixed b) Static
 c) Dynamic d) None
- Q.4 Gross errors arise due to
 a) Carelessness b) Poor Judgement
 c) Negligence d) All of above
- Q.5 LED emit light
 a) Only in red colour
 b) Only in yellow colour
 c) Only in green colour
 d) In red, green, yellow and amber colour

(180)

(4) 181534/121534/031534
/073633

(1) 181534/121534/031534
/073633

- Q.6 In a strip chart recorder
 a) Biological quantity is measured
 b) Chemical quantity is measured
 c) Physical quantity is measured
 d) No quantity is measured
- Q.7 The calibration of a voltmeter can be carried out by using
 a) A frequency meter b) A potentiometer
 c) An ammeter d) A function generator
- Q.8 For an Op-amp with negative feedback, the output is
 a) Equal to the input
 b) Increased
 c) Fed back to the inverting input
 d) Fed back to the non-inverting input
- Q.9 A differential amplifier
 a) Is a part of an Op-amp
 b) Has one input and one output
 c) Has two outputs
 d) Both (a) and (b)
- Q.10 A device which is used to measure a physical variable is known as .
 a) Instrument b) Sensor
 c) Controller d) None

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is Accuracy of an instrument?
 Q.12 Why calibration is needed?
 Q.13 What is data logging?
 Q.14 Define absolute error.

- Q.15 Full form of LED.
 Q.16 Sketch the circuit diagram of Op-Amp as an differentiator.
 Q.17 What is damping coefficient?
 Q.18 What is the use of RS-232C.
 Q.19 Define adjustment of an instrument.
 Q.20 What is virtual ground effect?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is the difference between field calibration and lab calibration process?
 Q.22 Define following terms:
 a) Dead Time b) Precision
 c) Sensitivity d) Resolution
 d) Hysteresis
- Q.23 Why measurement is necessary, write down its significance.
 Q.24 What is calibration explain it with suitable example.
 Q.25 Write down the list of process variable along with their measurement unit.
 Q.26 How to classify the errors, explain them?
 Q.27 What is slew rate and input offset current?
 Q.28 Discuss the advantages and disadvantages of circular chart recorder.
 Q.29 What are the precautions should be taken during measurement process?
 Q.30 What are field buses and how they are useful in measurement?